Datasets and reconstructed final state

System	Year	Integrated luminosity	Primary dataset
PbPb	2023	$1.72\,\mathrm{nb}^{-1}$	`HIPhysicsRawPrime (HIRun2023A)`
PbPb	2024	$1.62\,\mathrm{nb}^{-1}$	`HIPhysicsRawPrime* (HIRun2024*)`
ppRef	2024	$465.7\,\mathrm{pb}^{-1}$	`PPRefDoubleMuon (Run2024J)`

- Center-of-mass energy: $\sqrt{s_{\mathrm{NN}}} = 5.36\,\mathrm{TeV}$.
- Decay channels: $X(3872),\ \psi(2S) \to J/\psi\,\pi^+\pi^-$, with $J/\psi \to \mu^+\mu^-$.

Muon & track pre-selection: ppRef

Muons

☐ Soft-muon criteria

☐ Acceptance region

$p_{\mathrm{T}}^{\mu} > 3.5\,\mathrm{GeV}/c$	$\lvert\eta^{\mu}\rvert < 1.2$
$p_{\mathrm{T}}^{\mu} > 5.47 - 1.89\lvert\eta^{\mu}\rvert$	$1.2 \leq \lvert\eta^{\mu}\rvert < 2.1$
$p_{\mathrm{T}}^{\mu} > 1.5\,\mathrm{GeV}/c$	$2.1 \leq \lvert\eta^{\mu}\rvert < 2.4$

Tracks

☐ Quality

- $\sigma_{p_{\mathrm{T}}}/p_{\mathrm{T}} < 0.1$
- Pixel + strip hits > 10
- Normalized track $\chi^2 < 0.18$

☐ Acceptance

- $p_{\mathrm{T}} > 0.5\,\mathrm{GeV}/c$
- $\lvert\eta\rvert < 2.4$

Candidate

☐ Vertex prob. $> 0.5\%$

☐ $7.5 < p_{\mathrm{T}} < 50\,\mathrm{GeV}/c$

☐ `Btrk1dR`< 0.6

$\Delta m = m - m_{J/\psi} - m_{\pi\pi}$

☐ $\lvert y\rvert < 2.4$

☐ `BQvalue`< 0.15

☐ $\Delta m < 0.15\,\mathrm{GeV}/c^2$

Di-muon system

☐ Opposite muon charges

☐ Vertex probability $> 1\%$

☐ $\lvert m_{\mu\mu} - m_{J/\psi}^{\mathrm{PDG}}\rvert < 0.15\,\mathrm{GeV}/c^2$

Muon & track pre-selection: PbPb

Muons

☐ Hybrid-soft muons
- `isGlobalMuon` and `isTrackerMuon`
- $D_{xy} < 0.3$ cm; $D_z < 20$ cm
- `nPixWMea`> 0; `nTrkWMea`> 5

☐ Acceptance region

$p_{\mathrm{T}}^{\mu} > 3.5\,\mathrm{GeV}/c$	$\lvert\eta^{\mu}\rvert < 1.2$
$p_{\mathrm{T}}^{\mu} > 5.47 - 1.89\lvert\eta^{\mu}\rvert$	$1.2 \leq \lvert\eta^{\mu}\rvert < 2.1$
$p_{\mathrm{T}}^{\mu} > 1.5\,\mathrm{GeV}/c$	$2.1 \leq \lvert\eta^{\mu}\rvert < 2.4$

Tracks

☐ Quality
- $\sigma_{p_{\mathrm{T}}}/p_{\mathrm{T}} < 0.1$
- Pixel + strip hits > 10
- Normalized track $\chi^2 < 0.18$

☐ Acceptance
- $p_{\mathrm{T}} > 0.9\,\mathrm{GeV}/c$
- $\lvert\eta\rvert < 2.4$

Candidate

☐ Vertex prob. $> 0.5\%$
☐ $7.5 < p_{\mathrm{T}} < 50\,\mathrm{GeV}/c$
☐ `Btrk1dR`< 0.6
☐ $\lvert y\rvert < 2.4$
☐ `BQvalue`< 0.15
☐ $\Delta m < 0.15\,\mathrm{GeV}/c^2$

$\Delta m = m - m_{J/\psi} - m_{\pi\pi}$

Di-muon system

☐ Opposite muon charges
☐ Vertex probability $> 1\%$
☐ $\lvert m_{\mu\mu} - m_{J/\psi}^{\mathrm{PDG}}\rvert < 0.15\,\mathrm{GeV}/c^2$